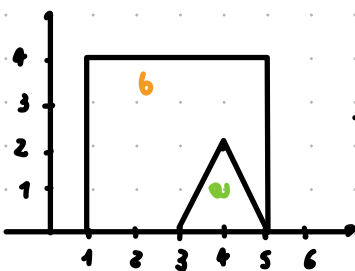
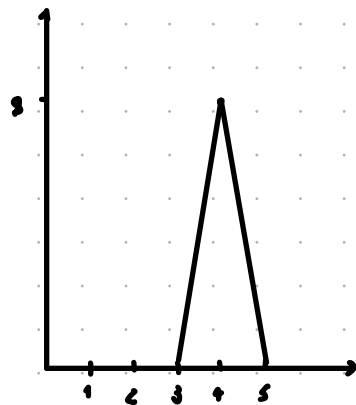


$$S(\ell) = 2\text{TRIANG}\left(\frac{\ell-4}{2}\right) + \text{RECT}\left(\frac{\ell-3}{4}\right)$$



$S(\ell)$



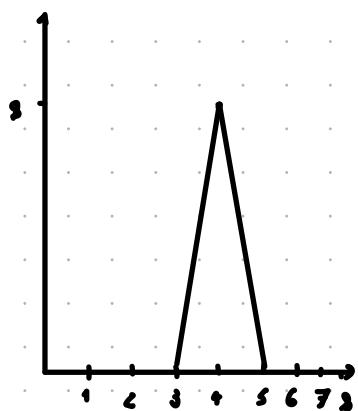
$$S(\ell) = 8\text{TRIANG}\left(\frac{\ell-4}{2}\right)$$

$$S(f) = 8 \cdot \frac{2}{2} \text{sinc}^2\left(f \frac{1}{2}\right) e^{-j2\pi f 4}$$

$$E = \int_{-\infty}^{\infty} S(\ell)^2 d\ell = 2 \int_0^1 64 \ell^2 d\ell = 128 \left[\frac{\ell^3}{3} \right]_0^1 = \frac{128}{3}$$



$$y = \omega x^2 \rightarrow 64 = \omega \cdot 1$$



$$T_0 = 8$$

$$C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} S(\ell) e^{-j2\pi n \ell / T_0} d\ell \rightarrow C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} S(\ell) e^{-j2\pi n \ell / T_0} d\ell = \frac{1}{8} \int_{-4}^{4} S(\ell) d\ell$$

$$C_n = \frac{1}{T_0} S\left(\frac{n}{T_0}\right) \rightarrow C_1 = \frac{1}{8} S\left(\frac{1}{8}\right) = \frac{1}{8} \cdot 8 \cdot \frac{2}{2} \text{sinc}^2\left(\frac{1}{8} \cdot \frac{2}{2}\right) e^{-j2\pi 1 f 4}$$

$$S(f) = 8 \cdot \frac{2}{2} \text{sinc}^2\left(f \frac{1}{2}\right) e^{-j2\pi f 4}$$